**12** Spheres X and Y form an isolated system. The mass of Y is greater than the mass of X.

Sphere Y is initially stationary.

Sphere X collides elastically with sphere Y.

The speed of sphere X before the collision is $u$.

Which statement **must** be correct?

**A** Sphere X rebounds with a speed that is greater than $u$, and sphere Y moves off with a speed that is less than $u$.

**B** Sphere X rebounds with a speed that is less than $u$, and sphere Y moves off with a speed that is also less than $u$.

**C** Sphere X rebounds with speed $u$, and sphere Y remains stationary.

**D** Sphere X remains stationary, and sphere Y moves off with a speed that is less than $u$.